1 Fig. 1.1 shows a coil made from resistance wire.

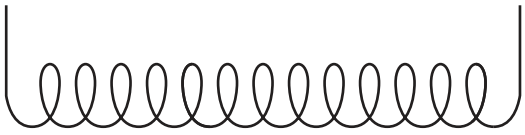

Fig. 1.1

The coil is placed in cooking oil of mass m. The total length of the resistance wire in the oil is L.

A potential difference V is applied to the coil. The temperature of the oil increases by $\Delta\theta$ in time t.

It is suggested that $\Delta\theta$ is related to L by the relationship

$$\frac{AtV^2}{L} = mK\Delta\theta + Z$$

where A is the cross-sectional area of the wire, and K and Z are constants.

Plan a laboratory experiment to test the relationship between $\Delta\theta$ and L.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and Z.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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